Supplementary Table 2.

To find those polymorphisms which affect the expression of many messages, we divided the genome into 20-kb bins and counted the number of messages linking to each bin (after removing *cis*-linkages). We empirically computed a significance threshold of 12 messages per bin: random assignment of linkages to bins produced bin heights this high in only 1% of simulations, i.e. all bins with more than 12 messages are significant at $p < 0.01$. Twenty-two bins fit this criterion. We combined messages from adjacent bins to form unified groups for the analysis below: Group 2 contains linkages from four bins, Group 11 from three bins, and Groups 3, 7, 8, and 12 from two bins each. Annotated messages in most groups allowed hypotheses about function and, in many cases, about the identity of polymorphic regulators controlling them. The results are listed in Table S2.

Group 2 contains the cluster of 12 genes shown in Figure 1 of the main text, whose regulator we confirmed to be Amn1. Most of these genes were previously shown to exhibit daughter-specific expression (DSE) or to be co-expressed with DSE genes in culture-averaged microarray experiments (1). Group 2 also contains 6 more DSE-related genes (**1**) and 11 which regulate or act directly in mitosis. This suggests more widespread influence by Amn1 on these functions than has been described thus far (**2**).

Group 2 also contains 46 genes known to be involved in synthesis or assembly of the ribosome. We infer the presence of another regulator, near but unrelated to *AMN1*, which controls these messages. A good candidate is the RNA helicase *MAK5*, implicated in

ribosomal biogenesis. We sequenced this gene and found five coding changes in RM relative to BY (G51A, Q239R, N328D, S576C, and S613P), plus a four amino-acid deletion starting at position 139.

Multiple additional regulators may control subsets of the 197 Group 2 genes which are not DSE- or ribosome-related.

Group 4 contains 100 messages linking to a region near the gene *LEU2* (an auxotrophic marker deleted in the RM strain). Of the Group 4 genes, 28 are involved in the biosynthesis of a wide range of amino acids, not just leucine and its close chemical relatives. This implies either the presence of another, more general regulator of amino acid catabolism in the chromosome III region--the amino acid permease *AGP1* may be a candidate--or a feedback mechanism caused by the *LEU2* deletion which influences amino acid biosynthesis across the board. Thirteen other messages in Group 4 are active in mitochondria or implicated in carbohydrate metabolism, suggesting that another polymorphic regulator, unrelated to *LEU2*, may lie in the region.

The 36 messages of Group 5 map to the *MAT* locus. Eighteen of the Group 5 genes have been implicated in mating or are known to be regulated by mating type.

Group 6 contains 19 messages linking to the *URA3* locus (another auxotrophic marker deleted in the RM strain). Of these 19, four code for proteins known to function in the uracil biosynthetic pathway.

Group 7 contains the cluster of 7 pheromone response genes shown in Figure 1 of the main text, whose regulator we confirmed to be *GPA1*. Another 12 of the Group 7 genes are involved in pheromone response or mating, and as such are good candidates for additional *GPA1* targets.

There are 114 messages in Group 8, which link to a region on chromosome XII containing the heme-dependent transcription factor *HAP1*. We showed previously that our BY strain carries a Ty insertion in *HAP1* (**3**). Of the 114 Group 8 genes, 28 are known to play a role in sterol or fatty acid biosynthesis, and 10 more are cytochromes, contain a heme cofactor, or are involved in heme biosynthesis.

Group 9 is highly enriched in messages whose genes lie near telomeres. We computed the start site positions of these genes in units of their respective chromosome lengths, and found that all but 2 of the 34 genes in Group 9 are located in the first or last 2% of chromosomes. The messages in Group 9 link to a region on chromosome XII containing *SIR3*. In contrast to results with a smaller cross (**3**), Group 9 contains other subtelomeric protein types beside Y' helicases.

Of the 13 messages in Group 13, five are subunits of the ATP synthase and five more have been implicated in respiration or mitochondrial function. A good candidate in the linkage region is *CAT5*, part of the coenzyme Q biosynthetic machinery.

Messages in the remaining groups did not share enough common function for us to make predictions about their regulators. Group 1 contains 6 ribosomal proteins, 2 RNA helicases, and a subunit of RNA pol III. Among the Group 10 messages are four known to be regulated by zinc and 11 that participate in amino acid or adenine catabolism. Group 11 contains 380 messages; the gene products of 119 are known to be active in or localize to mitochondria (**4**). Across the messages of Group 12, 57 were found to vary in expression in strains deleted for the stress response factors *MSN2/4* (**5**, **6**).

References:

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2. Wang Y, Shirogane T, Liu D, Harper JW, Elledge SJ. Exit from exit: resetting the cell cycle through Amn1 inhibition of G protein signaling. *Cell* 112(5):697-709. 2003.
3. Brem RB, Yvert G, Clinton R, Kruglyak L. Genetic dissection of transcriptional regulation in budding yeast. *Science* 296(5568):752-5. 2002.
4. Ashburner, M. *et al.* Gene ontology: tool for the unification of biology. The Gene Ontology Consortium. *Nat Genet* 25:25-9. 2000.
5. Rep M, Krantz M, Thevelein JM, Hohmann S. The transcriptional response of Saccharomyces cerevisiae to osmotic shock. Hot1p and Msn2p/Msn4p are required for the induction of subsets of high osmolarity glycerol pathway-dependent genes. *J Biol Chem* 275(12):8290-300. 2000.
6. Causton HC et al. Remodeling of yeast genome expression in response to environmental changes. *Mol Biol Cell* 12(2):323-37. 2001.

Summary of groups of messages linking to loci with widespread transcriptional effects.

Group	# msgs	Common function	Linkage bin	Putative regulator	% msgs from Brem et al. recapitulated
1	26	Ribo proteins	II:390000	Unknown	N/A
2	272	DSE rRNA processing	II:560000	*AMN1* *MAK5*	94
3	14	Unknown	II:710000	Unknown	N/A
4	100	AA catabolism	III:100000	*LEU2*	95
5	36	Mating	III:230000	*MAT*	89
6	19	Ura catabolism	V:130000	*URA3*	71
7	62	Phero response	VIII:130000	*GPA1*	N/A
8	114	Heme/FA	XII:680000	*HAP1*	89
9	34	Subtelomeric	XII:1070000	*SIR3*	94
10	21	Unknown	XIII:70000	Unknown	N/A
11	380	Mitochondria	XIV:503000	Unknown	91
12	174	Msn2/4 targets	XV:180000	Unknown	74
13	13	Respiration	XV:590000	*CAT5*	N/A

Groups of messages linking to loci with widespread transcriptional effects.

Group 1

YIL045W PIG2 Protein that interacts with Gsy2p, possible regulatory subunit for the PP1-family protein phosphatase Glc7p
YER110C KAP123 Karyopherin beta 4
YDL083C RPS16B Ribosomal protein S16B (rp61R)
YPL143W RPL33A Ribosomal protein L33A (L37A) (YL37) (rp47)
YIL133C rpl16A Ribosomal protein L16A (L21A) (rp22) (YL15)
YOL040C RPS15 40S ribosomal protein S15 (S21) (rp52) (RIG protein)
YPL220W RPL1A Ribosomal protein L1A, forms part of the 60S ribosomal subunit
YJR118C ILM1 Protein of unknown function, possibly involved in mitochondrial DNA maintenance
YER009W NTF2 nuclear transport factor, homologous to mammalian cytosolic nuclear import factor NTF2
YCR043C
YDR321W asp1 Asparaginase I, intracellular isozyme
YJL190C RPS22A Ribosomal protein S22A (S24A) (rp50) (YS22)
YKR096W
YDR186C
YOL116W MSN1 43 kDa protein, transcriptional activator
YDL102W cdc2 largest and catalytic subunit of DNA polymerase III (delta)
YHR159W
YKR059W tif1 Translation initiation factor 4A (eIF4A), member of the DEAD-box family of ATP-dependent RNA helicases, identical to Tif2p
YLR017W MEU1 regulator of ADH2 expression
YGR123C PPT1 serine\/threonine phosphatase
YOR207C ret1 second-largest subunit of RNA polymerase III
YOR098C nup1 nuclear pore complex protein
YNL173C MDG1 Involved in G-protein mediated signal transduction
YGR130C
YIL095W PRK1 probable serine\/threonine-protein kinase
YGL120C PRP43 Pre-mRNA splicing factor required for release of the lariat-intron from the spliceosome, member of the DEAH-box RNA helicase family

Group 2

YJR073C OPI3 Methylene-fatty-acyl-phospholipid synthase (unsaturated phospholipid N-methyltransferase)
YLR249W yef3 EF-3 (translational elongation factor 3)
YER034W
YGL075C MPS2 nuclear envelope\/ER protein involved in chromosomal segregation
YHR185C PFS1 Protein required for prospore membrane formation during sporulation
YKR026C gcn3 34 KD alpha subunit of eIF2B

YDR285W ZIP1 synaptonemal complex protein
YLR063W
YMR191W
YLL034C Member of the AAA family of ATPases required for 60S ribosomal biogenesis
YGR173W
YDR083W RRP8 Protein involved in cleavage at site A2 in pre-rRNA in the pathway of ribosomal RNA processing
YPR075C OPY2 Protein that may be involved in cell-cycle regulation, overproduction causes insensitivity to alpha-factor arrest
YOR146W
YOR340C RPA43 DNA-dependent RNA polymerase I subunit A43
YKR081C RPF2 Protein required for large ribosomal subunit assembly and maturation of precursor rRNAs
YHL032C GUT1 glyerol kinase (converts glycerol to glycerol-3-phosphate
YKL172W EBP2 Protein involved in pre-rRNA processing and ribosomal subunit assembly
YER006W NUG1 Nuclear protein required for 60S ribosomal subunit export from the nucleus, has similarity to mouse Gna-rs1 protein, which is a putative GTP-binding protein
YMR291W Serine/threonine protein kinase of unknown function
YMR185W
YNL157W
YOR270C vph1 vacuolar ATPase V0 domain subunit a (100 kDa)
YIL104C SHQ1 Nucleoplasmic protein that forms a complex with Naf1p, required for the biogenesis of box H/ACA snoRNPs
YPL055C LGE1 Protein possibly involved in DNA repair
YIL106W MOB1 Protein required for completion of mitosis and maintenance of ploidy
YLR350W ORM2 Putative transmembrane protein that may function in protein folding in the endoplasmic reticulum and in the cell wall integrity pathway
YGL016W KAP122 Nuclear transport factor, member of karyopherin-beta family
YAL035W YIF2 Translation initiation factor 2, has similarity to bacterial IF2
YHR017W YSC83 similar to S. douglasii YSD83
YMR139W RIM11 Serine\/threonine protein kinase, phosphorylates the mitotic activator IME1
YMR103C
YOL123W HRP1 putative polyadenylated-RNA-binding protein
YOR233W KIN4 protein kinase
YDR211W GCD6 Translation initiation factor eIF-2B epsilon subunit
YDL027C
YNL112W DBP2 ATP-dependent RNA helicase of DEAD box family
YER036C KRE30 Member of the non-transporter group of the ATP-binding cassette (ABC) superfamily
YGR200C ELP2 90 kD subunit of elongator and elongating RNA pol II holoenzyme
YNL068C FKH2 Homolog of Drosophila forkhead protein, involved in transcriptional silencing, cell morphology and cell cycle
YKR060W

YHR081W LRP1 Protein involved in nonhomologous end-joining and homologous recombination
YKL082C Nucleolar component of pre-60S ribosomal particle
YMR200W ROT1 Calmodulin binding; Phosphatidylinositol binding; Cell wall organization and biogenesis
YLR197W SIK1 Nucleolar protein component of box C/D snoRNPs, which are necessary for 2'-O-methylation of ribosomal RNAs, component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis
YNL170W
YOR195W SLK19 Protein involved with Kar3p in control of spindle dynamics, cleaved by Esp1p (separase)
YCL008C stp22 Protein required for vacuolar targeting of temperature-sensitive plasma membrane proteins such as Ste2p and Can1p, component of endosomal sorting complex required for transport (ESCRT-1) with Vps28p and Srn2p
YDR479C
YOL110W SHR5 involved in RAS localization and palmitoylation
YNL102W pol1 DNA polymerase I alpha subunit, p180
YNL188W kar1 Component of spindle pole body half-bridge, required for spindle pole body duplication and karyogamy
YDL209C CWC2 Putative spliceosomal protein with similarity to S. pombe cwf2p
YDL214C PRR2 Serine/threonine protein kinase potentially involved in pheromone response
YER152C
YGR089W
YOL006C top1 topoisomerase I
YKR023W
YGR116W SPT6 transcriptional regulator, interacts with histones, primarily histone H3, possesses nucleosome assembly activity
YMR173W-A
YMR173W DDR48 flocculent specific protein\; contains >35 repeats of the amino acid sequence NNNDSYGS
YLR287C
YOR263C
YOR264W DSE3
YGR041W BUD9 involved in bud-site selection
YPR106W ISR1 protein kinase
YKL132C RMA1 probable folyl-polyglutamate synthetase
YHR143W DSE2 Ser-Thr rich protein
YGL028C SCW11 soluble cell wall protein
YER124C DSE1
YJL078C PRY3 Similar to plant PR-1 class of pathogen related proteins
YNL327W EGT2 Cell wall protein. Putatively involved in glucan metabolism
YNL066W SUN4 Protein involved in the aging process
YNR067C DSE4
YPR105C COD1 Component of the Cog3p-Cog2p complex involved in vesicular transport to the Golgi

YMR285C NGL2 Protein required for 5.8 S rRNA processing, has possible DNAse/RNAse function, member of the magnesium-dependent ExoIII-APE nuclease family
YPL069C BTS1 geranylgeranyl diphosphate synthase
YLL032C
YDL055C PSA1 mannose-1-phosphate guanyltransferase; mutants exhibit failure of cell separation and/or cytokinesis, impaired bud growth and bud-site selection (Warit et al.)
YNR056C BIO5 transmembrane regulator of KAPA\/DAPA transport
YDR161W TCI1 interacts with PP2C
YPL094C SEC62 membrane component of ER protein translocation apparatus
YJL072C
YDR119W Member of the 14-spanner drug:[H+] antiporter (DHA14) family of multidrug-resistance (MFS-MDR) proteins in the major facilitator superfamily (MFS)
YOR220W
YCR094W CDC50 involved in cell cycle
YHR023W MYO1 Class II Myosin
YDR443C SRB9 Component of RNA polymerase II holoenzyme and Kornberg's mediator (SRB) subcomplex
YLR323C
YOR298C-A MBF1 multiprotein bridging factor, transcriptional coactivator of Gcn4p
YGR276C RNH70 ribonuclease H
YGR080W TWF1 Twinfilin A, an actin monomer sequestering protein
YGL087C MMS2 Similar to ubiquitin conjugating protein family
YLR370C ARC18 Arp2\/3 Complex Subunit
YJR147W HMS2 Transcription factor with a probable role in pseudohyphal growth
YLR042C
YGR014W msb2 Osmosensor protein, functions in parallel and partially redundant to Sho1p in activating the Ste11p branch of the HOG pathway
YGR161C
YOR342C
YPL081W RPS9A Ribosomal protein S9A (S13) (rp21) (YS11)
YBR189W RPS9B Ribosomal protein S9B (S13) (rp21) (YS11)
YNL011C
YLR285W
YPR183W DPM1 dolichol phosphate mannose synthase
YKR028W SAP190 Type 2A-related protein phosphatase
YJL013C MAD3 spindle checkpoint complex subunit
YDR169C STB3 Sin3 binding protein
YOL032W
YML029W
YLR400W
YJL010C
YOR168W gln4 glutaminyl-tRNA synthetase
YMR290C HAS1 Putative RNA-dependent helicase
YJR041C
YPR010C RPA135 RNA polymerase I subunit A135

YPL146C Nuclear and nucleolar protein associated with the nuclear pore complex
YLR276C DBP9 Member of the DEAD-box RNA helicase family, functions in rRNA processing to the precursor of 60S ribosomal subunits, interacts with Dbp6p
YGR187C HGH1 (putative) Hmg1\/2 protein
YPL086C ELP3 RNA polymerase II-associated Histone acetyltransferase
YER149C PEA2 Protein involved in oriented growth toward mating partner
YOR308C SNU66 66kD U4\/U6.U5 snRNP associated protein
YMR290W-A
YOR047C STD1 homologous to MTH1\; interacts with the SNF1 protein kinase and TBP in two-hybrid and in in vitro binding studies
YNL299W TRF5 exhibits homology to Trf4p and Top1p
YGR140W CBF2 110 kDa subunit of the centromere binding factor CBF3
YBL004W
YBR034C HMT1 nuclear protein arginine methyltransferase (mono- and asymmetrically dimethylating enzyme)
YKR024C DBP7 Protein involved in 60S ribosomal large subunit biogenesis, member of DEAD-box family of putative ATP-dependent RNA helicases
YIL091C
YNR053C NOG2 Nuclear protein involved in export of the 60S ribosomal subunit from the nucleus
YJR002W MPP10 U3 snoRNP protein
YNL182C IPI3 Protein containing three WD domains (WD-40 repeat)
YLR222C UTP13 Component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis, overproduction causes chromosome instability and increased mitotic recombination, contains WD (WD-40) repeats
YJL098W SAP185 Sit4p-associated protein
YLR336C SGD1 may be involved in high osmolarity signaling pathway
YDL148C NOP14
YDR324C UTP4 Protein with multiple WD40 repeats, component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis
YOR154W
YLR143W
YNL227C Member of the DnaJ family of protein chaperones
YIL019W
YPR136C FYV15
YPR076W
YGR007W MUQ1 choline phosphate cytidylyltransferase (also called phosphoethanolamine cytidylyltransferase or phosphocholine cytidylyltransferase)
YNL159C ASI2 Protein involved in regulation of amino acid permease gene expression
YIL108W
YDR411C
YPR154W PIN3
YNL113W rpc19 subunit common to RNA polymerases I (A) and III (C)
YMR141C
YDR101C
YKL154W SRP102 Signal recognition particle receptor beta subunit

YBL028C
YHR069C RRP4 3->5 exoribonuclease\; Component of the exosome 3->5 exonuclease complex with Rrp41p, Rrp42p, Rrp43p and Dis3p (Rrp44p).
YOL124C
YKL078W DHR2 RNA helicase of DEAH box family, involved in ribosomal RNA maturation
YPL126W NAN1 Nucleolar protein associated with Net1p, component of U3 snoRNP (also called small subunit processome), which is required for 18S biogenesis
YHR170W NMD3 cytoplasmic factor required for a late cytoplasmic assembly step of the 60S subunit
YJL033W HCA4 Probable RNA helicase CA4, involved in ribosomal RNA processing
YGL078C DBP3 ATP-dependent RNA helicase CA3 of the DEAD\/DEAH box family
YDL060W TSR1 Protein required for rRNA processing and 40S ribosomal subunit biogenesis
YLR401C DUS3 Member of dihydrouridine synthase (Dus) family (Dus1p, Smm1p/Dus2p, Ylr401cp, Ylr405wp), may effect dihydrouridine modification of tRNA
YER082C UTP7 Component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis
YKL143W LTV1 Protein that may play a role in pre-rRNA processing, required for viability at low temperature
YOL144W NOP8 Nucleolar protein required for 60S ribosome biogenesis; rRNA processing
YJR008W
YPR112C MRD1 Protein with similarity to Pab1p, Pub1p, Nsr1p, Nop4p and other RNA-binding proteins, contains multiple RNA-binding domains, is required for 35S rRNA processing
YNL022C
YPL207W Member of the flavodoxin family of electron-transfer proteins
YGR090W UTP22 Member of the nucleolar RNA-associated proteins (Nrap) protein family
YCR057C PWP2 Essential protein required for cell separation, has eight WD (WD-40) repeats, component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis
YLR051C
YDR496C PUF6
YIL096C
YDL153C SAS10 Protein that derepresses HMR, HML and telomeres when overproduced, component of U3 snoRNP (also called small subunit processome), which is required for 18S biogenesis
YHR065C RRP3 RRP3 is a DEAD box gene homologous to eIF-4a which encodes an RNA-dependent ATPase possessing helicase activity which is not specific for RNA
YFL023W BUD27 Protein that may be involved in bud site selection and bipolar budding
YDR021W FAL1 DEAD-box protein, putative RNA helicase

YHR088W RPF1 Protein required for the maturation of the large ribosomal subunit precursor rRNAs, member of the Imp4p superfamily of proteins with a sigma70-like motif
YCL059C KRR1 involved in cell division and spore germination
YGL029W CGR1 Predicted coiled-coil protein required required for 27S pre-rRNA processing and 60S ribosomal subunit biogenesis, down-regulated after diauxic shift
YNL308C KRI1 Protein required for synthesis of 40S ribosomal subunits, binds Krr1p and has weak similarity to Mgm1p
YKL083W
YGR145W ENP2 Component of NuA3 histone acetyltransferase complex
YGL171W ROK1 RNA helicase involved in rRNA processing
YPL043W NOP4 Nucleolar protein required for ribosome biogenesis
YOL041C NOP12 Protein important for the synthesis of 25S pre-rRNA
YJL148W RPA34 unshared RNA polymerase I subunit
YCR016W
YNL110C
YDR312W SSF2 possibly involved in mating
YDR060W MAK21 Protein required for 60S ribosomal subunit biogenesis
YKL009W MRT4 involved in mRNA decay
YHR052W CIC1 Protein associated with the 26S proteasome, may be involved in regulated degradation of F-box proteins Cdc4p and Grr1p
YAL025C mak16 Response to viruses; Ribosomal large subunit biogenesis
YMR239C RNT1 Double-strand-specific ribonuclease required for rRNA processing, homolog of E. coli RNase III
YGR280C
YNR054C Protein with similarity to mouse ABT1, an activator of RNA Polymerase II basal transcription
YDR299W BFR2 involved in secretion
YOR287C
YOR204W DED1 ATP-dependent RNA helicase of DEAD box family
YER039C HVG1 (putative) nucleotide sugar transporter
YDR436W PPZ2 serine-threonine phosphatase Z
YLR106C
YIL077C
YLR375W STP3 Involved in pre-tRNA splicing
YDR525W-A SNA2 Protein involved in Prc1p, Pep4p, and Pho8p trafficking to the vacuole
YIL042C Protein with similarity to mitochondrial branched chain alpha-ketoacid and pyruvate dehydrogenase protein kinases
YNL208W
YDR043C NRG1 transcriptional repressor which can bind to UAS-1 in the STA1 promoter and which can interact with Ssn6p
YDR255C
YDR171W HSP42 heat shock protein similar to HSP26, involved in cytoskeleton assembly

YDL106C pho2 Homeobox-domain containing transcription fractor which is a positive regulator of PHO5 and other genes.
YIR016W
YJL079C PRY1 Similar to plant PR-1 class of pathogen related proteins
YOR243C PUS7 Pseudouridine synthase required for pseudouridine modification of U35 of U2 snRNA
YJL050W MTR4 Protein required for mRNA export from nucleus, member of the DEAD-box RNA helicase family
YPL247C
YFR017C
YJL141C YAK1 Serine-threonine protein kinase
YBR214W SDS24 nuclear protein similar to pombe sds23
YMR181C
YDL023C
YGR177C ATF2 Alcohol acetyltransferase
YMR274C RCE1 protease, acts on Ras and a-factor C-termini
YML101C
YAR007C RFA1 69 kDa subunit of the heterotrimeric RPA (RF-A) single-stranded DNA binding protein, binds URS1 and CAR1
YNL219C ALG9 mannosyltransferase
YIL153W RRD1 Resistant to Rapamycin Deletion
YNL241C zwf1 Glucose-6-phosphate dehydrogenase
YDL123W
YHR199C
YIL154C IMP2' transcription factor
YER018C SPC25 component of spindle pole
YGL126W SCS3 involved in inositol biosynthesis
YCL044C
YLR352W
YPL258C THI21 THI for thiamine metabolism. Transcribed in the presence of low level of thiamine (10-8M) and turned off in the presence of high level (10-6M) of thiamine. Under the positive control of THI2 and THI3.
YFR005C SAD1 Protein required for pre-mRNA splicing and for snRNP assembly
YDL003W MCD1 involved in mitosis, similar to pombe Rad21
YMR184W
YOR288C MPD1 Disulfide isomerase related protein
YGL127C SOH1 SOH1 encodes a novel 14-kD protein with limited sequence similarity to RNA polymerases. The Soh1 protein interacts with a DNA repair protein, Rad5p, in a two-hybrid system assay.
YOL031C Protein involved in protein translocation into endoplasmic reticulum, associated with Kar2p and Sec63p
YOR151C RPB2 second largest subunit of RNA polymerase II
YLR099C ICT1 Increased Copper Tolerance\; Similar to Ecm18p
YMR030W
YDL079C MRK1 MDS1 related protein kinase
YCL036W

YPL158C
YPR133C IWS1 Protein that may be involved in transcription elongation
YMR132C
YDR156W RPA14 RNA polymerase I subunit A14
YAL058W CNE1 Calnexin and calreticulin homolog
YAL014C
YGR129W SYF2 (putative) involved in pre-mRNA splicing
YGL246C RAI1
YNL078W JIP1 localizes to the bud neck at the time of cell division, suggesting a role in cell division
YKL185W ASH1 probable purine nucleotide-binding protein
YGR098C esp1 involved in regulation of spindle pole body duplication
YGR163W GTR2 (putative) small GTPase, similar to Gtr1
YLR372W SUR4 involved in fatty acid biosynthesis
YDR536W STL1 sugar transporter-like protein
YHR153C SPO16 sporulation protein
YGR109C CLB6 B-type cyclin
YIL009W FAA3 Acyl CoA synthase
YDR149C
YKL014C
YAL020C ATS1 similarity to human RCC1 protein
YGR257C Member of the mitochondrial carrier family (MCF) of membrane transporters
YLR079W SIC1 Cdc28p-Clb5 protein kinase inhibitor
YPL157W TGS1 Conserved S-adenosylmethionine-dependent methyltransferase, required for the trimethylguanosine (TMG) structure of snRNAs and snoRNAs, localized to the nucleolus

Group 3

YML043C RRN11 component of rDNA transcription factor
YJL129C trk1 180 kDa high affinity potassium transporter
YJL053W PEP8 Vacuolar protein similar to mouse gene H<beta>58; Integral to vacuolar membrane
YMR222C
YLR280C
YGL013C PDR1 zinc-finger transcription factor of the Zn(2)-Cys(6) binuclear cluster domain type; Sphingolipid biosynthesis; Iron ion transport
YIR004W DJP1 involved in peroxisome biogenesis
YDL167C NRP1 Asparagine-rich protein with one RNA recognition (RRM) domain
YNL322C kre1 Cell wall protein needed for cell wall beta-1,6-glucan assembly, probably involved in side-chain addition to form mature beta-1,6-glucan, plasma membrane receptor for K1 killer toxin
YKL050C
YKL019W RAM2 CAAX farnesyltransferase alpha subunit
YIL130W GIN1 Protein that contains a Zn[2]-Cys[6] fungal-type binuclear cluster domain in the N-terminal region, putative transcription factor

YDR519W fkb2 FKBP (FK506 binding protein) 13\; peptidylprolyl cis-trans isomerase activity; Integral to membrane; Endoplasmic reticulum
YCR073W-A SOL2 Possible 6-phosphogluconolactonase, involved in tRNA synthesis and function

Group 4

YNR027W BUD17
YPR059C
YKL030W
YKL029C MAE1 mitochondrial malic enzyme
YOR271C Member of the mitochondrial tricarboxylate carrier family of membrane transporters, has similarity to rat tricarboxylate carrier
YEL071W DLD3 D-lactate dehydrogenase
YCR005C CIT2 non-mitochondrial citrate synthase
YGL184C STR3 Cystathionine beta-lyase
YGR267C fol2 GTP cyclohydrolase I, catalyzes synthesis of D-erythro-7,8-dihydroneopterintriphosphate from GTP in biosynthesis of tetrahydrobiopterin
YOR202W his3 imidazoleglycerol-phosphate dehydratase
YMR062C ARG7 Ornithine acetyltransferase, catalyzes the fifth step in ornithine and arginine biosynthesis, also has acetylglutamate synthase activity
YDL057W
YKL211C trp3 anthranilate synthase Component II and indole-3-phosphate (multifunctional enzyme)
YOL064C MET22 3'(2')5'-bisphosphate nucleotidase
YDR158W hom2 Aspartate-semialdehyde dehydrogenase (L-aspartate-beta-semialdehyde:NADP oxidoreductase [phosphorylating]), second step in common pathway for methionine and threonine biosynthesis
YNL036W NCE103 involved in non-classical protein export pathway; Protein involved in protection against oxidative damage
YBR047W
YOR221C MCT1 malonyl-CoA:ACP transferase
YGL114W Protein member of the oligopeptide transporter (OPT) family of membrane transporters
YGL186C TPN1 Member of the purine/cytosine permease family of membrane transporters
YKL001C met14 adenylylsulfate kinase
YNL277W met2 homoserine O-trans-acetylase
YJR137C ECM17 Putative sulfite reductase
YFR030W met10 subunit of assimilatory sulfite reductase
YJL212C OPT1 Oligopeptide transporter\; Opt1p transports tetra- and pentapeptides, including the endogenous opioid pentapeptide leucine enkephalin.
YDR481C pho8 repressible alkaline phosphatase
YIR017C MET28 transcriptional activator in the Cbf1p-Met4p-Met28p complex
YOR292C Member of the Mpv17 and PMP22 family, which are peroxisomal proteins
YIL074C SER33 3-phosphoglycerate dehydrogenase

YER081W SER3 3-phosphoglycerate dehydrogenase
YOL081W ira2 encodes a GTPase activating protein, highly homologous to Ira1p, homologue of neurofibromin
YLR319C BUD6 Actin Interacting Protein
YLR113W HOG1 mitogen-activated protein kinase (MAP kinase)
YGL010W
YOR226C ISU2 Iron-sulfur cluster nifU-like protein
YOR225W
YPL181W CTI6 Protein that physically links the SAGA complex to the Ssn6p-Tup1p repressor complex, acts in concert with the SAGA complex to relieve Ssn6p-Tup1p mediated transcriptional repression
YER098W UBP9 ubiquitin carboxyl-terminal hydrolase
YOR375C GDH1 NADP-specific glutamate dehydrogenase
YAL062W GDH3 NADP-linked glutamate dehydrogenase
YDL066W IDP1 Mitochondrial form of NADP-specific isocitrate dehydrogenase
YOR184W ser1 phosphoserine transaminase
YCL030C his4 Phosphoribosyl-AMP cyclohydrolase / phosphoribosyl-ATP pyrophosphohydrolase / histidinol dehydrogenase, second, third, and tenth steps of histidine biosynthesis pathway
YBR104W YMC2 mitochondrial carrier protein
YIL003W DRE3 Essential mitochondrial protein with a nucleotide (ATP,GTP) binding motif
YML093W UTP14 Component of U3 snoRNP (also called small subunit processome), which is required for 18S rRNA biogenesis
YKL120W OAC1 mitochondrial oxaloacetate transport protein
YGL009C leu1 isopropylmalate isomerase
YHR208W BAT1 branched-chain amino acid transaminase, highly similar to mammalian ECA39, which is regulated by the oncogene myc
YLR348C DIC1 mitochondrial dicarboxylate transport protein
YGL138C
YBR068C BAP2 probable amino acid permease for leucine, valine, and isoleucine
YPL269W KAR9 Component of cortical adaptor complex that orients microtubules
YOR108W LEU9 Alpha-isopropylmalate synthase II
YNL104C LEU4 alpha-isopropylmalate synthase (2-Isopropylmalate Synthase)
YER073W ALD5 mitochondrial Aldehyde Dehydrogenase
YMR108W ilv2 acetolactate synthase
YJR016C ilv3 dihydroxyacid dehydratase
YKR071C
YGR146C
YJR130C STR2 Cystathionine gamma-synthase
YER175C TMT1
YBR256C rib5 Riboflavin synthase alpha-chain
YOL119C MCH4 Member of the monocarboxylate porter (MCP) family of the major facilitator superfamily (MFS), has similarity to mammalian monocarboxylate transporter proteins
YIR034C lys1 saccharopine dehydrogenase

YJR109C CPA2 Carbamoylphosphate synthetase of arginine biosynthetic pathway, synthetase (large) subunit
YGL117W
YOR130C ORT1 amino acid transporter
YDL170W UGA3 zinc-finger transcription factor of the Zn(2)-Cys(6) binuclear cluster domain type
YJR111C
YKL218C SRY1 Pyridoxal-5'phosphate-dependent enzyme homologous to mouse glial serine racemase
YHR071W PCL5 G1\/S cyclin (weak)
YDR035W ARO3 DAHP synthase; Aromatic amino acid family biosynthesis
YDL198C YHM1 (putative) mitochondrial carrier protein
YBR145W ADH5 alcohol dehydrogenase isoenzyme V
YOL058W arg1 arginosuccinate synthetase; arginine biosynthesis
YGL062W PYC1 pyruvate carboxylase
YHR029C Thymidylate synthase (putative\; weak)
YMR096W SNZ1 encodes highly conserved 35 kDa protein that shows increased expression after entry into stationary phase
YMR095C SNO1 SNZ1 proximal ORF, stationary phase induced gene
YMR094W CTF13 58 kd component (Cbf3c) of the multisubunit 'Cbf3' kinetochore protein complex, which binds to the CDE III element of centromeres
YOL118C
YJL087C TRL1 tRNA ligase
YPL084W BRO1 BCK1-like resistance to osmotic shock
YER003C PMI40 mannose-6-phosphate isomerase
YGL059W
YPL135W ISU1 Iron-sulfur cluster nifU-like protein
YGL125W met13 putative methylenetetrahydrofolate reductase (mthfr)
YBR111C YSA1 Nucleoside diphosphate-sugar hydrolase of the MutT (nudix) family
YIL118W RHO3 ras homolog--GTP binding protein
YHR063C PAN5 Ketopantoate Reductase
YOR044W
YLR027C AAT2 aspartate aminotransferase, cytosolic; Glutamate metabolism
YDR080W vps41 component of vacuolar membrane protein complex
YPR167C MET16 3'phosphoadenylylsulfate reductase
YLR092W SUL2 high affinity sulfate permease
YJL012C VTC4 polyphosphate synthetase (putative)
YML123C pho84 inorganic phosphate transporter, transmembrane protein
YOR155C
YLR149C

Group 5

YGL033W HOP2 Meiosis-specific gene required for the pairing of homologous chromosomes
YGL032C aga2 adhesion subunit of a-agglutinin

YCR097W-A
YIL015W bar1 extracellular protease synthesized in a-cells that cleaves and inactivates alpha factor
YFL026W ste2 alpha-factor pheromone receptor\; seven-transmembrane domain protein
YKL209C ste6 ABC transporter, glycoprotein, component of a-factor secretory pathway
YCR097W a1 haploid specific gene repressor
YOR090C PTC5 Type 2C Protein Phosphatase
YLR041W
YLR040C in MAT expression cluster, Spellman et al
YKL178C STE3 a factor recptor
YCL065W
YCL066W HMLalpha1 transcription factor involved in the regulation of alpha-specific genes
YPL187W mf(alpha)1 mating factor alpha
YKL177W
YNL145W mfa2 mating a-factor pheromone precursor
YFL025C BST1 negative regulator of COPII vesicle formation
YOR162C YRR1 transcription factor
YJR004C sag1 alpha-agglutinin
YGL090W LIF1 interacts with DNA ligase protein
YGL089C mf(alpha)2 alpha mating factor
YCL067C HMLAlpha2 Mating type protein alpha-2
YCR096C a2 Regulatory protein MATa2p (no known function)\; sequence is the same as the last 119 residues of MATalpha2p
YJR017C ess1 Peptidyl-prolyl cis\/trans isomerase (PPIase)
YNL146W induced by mating pheromones, Erdman et al.
YPL083C SEN54 54kDa subunit of the tetrameric tRNA splicing endonuclease
YNL105W
YFL027C GYP8 induced 28-fold by alpha mating pheromone
YOL025W LAG2 involved in determination of longevity
YHR178W STB5 Zinc finger (6-Cys)
YJL170C ASG7 An a-specific gene that is induced to a higher expression level by alpha factor
YLR288C MEC3 DNA damage checkpoint protein
YDR461W mfa1 a-factor mating pheromone precursor
YOR055W
YIL017C VID28 vacuole import and degradation
YDR459C

Group 6

YKR082W NUP133 nuclear pore protein
YMR043W mcm1 transcription factor containing the 56 amino-acid MADS (MCM1, AG, DEFAm SRF)-box motif within its DNA binding domain, plays a central role in the formation of both repressor and activator complexes

YBR218C PYC2 pyruvate carboxylase
YLR089C Putative mitochondrial alanine aminotransferase
YBR105C VID24 peripheral vesicle membrane protein
YER088C DOT6 nuclear protein with Myb domain involved in telomeric silencing
YDR192C NUP42 42-kD protein associated with nuclear pore complexes\; Nup42p is structurally related to the FG-nucleoporin family of pore proteins
YIL031W ULP2
YOR067C alg8 glycosyl transferase
YOL126C MDH2 cytosolic malate dehydrogenase
YOL143C RIB4 6,7-dimethyl-8-ribityllumazine synthase (DMRL synthase)
YLR420W ura4 dihydrooratase
YKL216W ura1 dihydroorotate dehydrogenase
YLR421C RPN13 New 26S proteasome subunit
YMR241W YHM2 DNA-binding protein, mtDNA stabilizing protein, mitochondrial inner membrane protein with low homology to RIM2
YJL130C ura2 carbamoyl-phophate synthetase, aspartate transcarbamylase, and glutamine amidotransferase
YLR298C YHC1 U1 snRNP protein
YGL134W PCL10 PHO85 cyclin
YNL303W

Group 7

YJR054W
YDR265W PEX10 C3HC4 zinc-binding integral peroxisomal membrane protein
YMR001C cdc5 protein kinase which functions at the G(sub)2\/M boundary
YDR207C UME6 Ume6p is a C6 zinc finger URS1-binding protein.
YJL199C
YDR283C gcn2 eukaryotic initiation factor 2 alpha (eIF2-alpha) kinase
YOL107W
YJR062C NTA1 52-kDa amidase specific for N-terminal asparagine and glutamine
YOR180C DCI1 Peroxisomal delta(3,5)-delta(2,4)-dienoyl-CoA isomerase
YBL011W SCT1 suppressor of choline-transport mutants
YBR272C HSM3 Hsm3p may be a member of the yeast MutS homolog family
YJL113W The TyB Gag-Pol protein. Gag processing produces capsid proteins. Pol is cleaved to produce protease, reverse transcriptase, and integrase activities.
YIR023W dal81 Transcriptional activator for allantoin and GABA catabolic genes, contains a Zn[2]-Cys[6] fungal-type binuclear cluster domain in the N-terminal region
YGR069W
YDR249C
YAR014C BUD14 maximal growth
YDR503C LPP1 Lipid phosphate phosphatase
YGL060W
YGR092W dbf2 Serine\/threonine protein kinase
YKL104C GFA1 Glutamine_fructose-6-phosphate amidotransferase (glucoseamine-6-phosphate synthase)

YHR142W CHS7 The seventh gene identified that is involved in chitin synthesis\; involved in Chs3p export from the ER
YIL117C PRM5 Protein transcriptionally regulated by cell stress, the pheromone response pathway, and the cell integrity signaling pathway
YIL009C-A EST3 Telomerase component, produced in two forms by programmed +1 translational frameshifting
YOR066W
YEL032W mcm3 component of DNA replication initiator complex
YDR410C ste14 farnesyl cysteine-carboxyl methyltransferase
YNL258C DSL1 Protein involved in ER to Golgi transport and required for Golgi-to-ER retrograde transport
YIL080W Ty3B Gag-Pol protein. Gag processing produces capsid proteins. Pol is cleaved to produce protease, reverse transcriptase and integrase activities.
YJR153W PGU1 Endo-polygalacturonase
YBR083W tec1 transcription factor of the TEA\/ATTS DNA-binding domain family, regulator of Ty1 expression□
YBL016W FUS3 cdc2+\/CDC28 related kinase with positive role in conjugation
YKL189W HYM1 Protein required for sporulation, apical cell growth, and efficient cell separation
YDR085C afr1 Protein involved in morphogenesis of the mating projection (shmoo)
YPR115W
YFR026C
YDR124W
YHL006C SHU1 suppressor of HU sensitivity of sgs1 mutations
YPL163C SVS1 Serine and threonine rich protein.
YHR084W ste12 Transcription factor that binds to pheromone response element (PRE) to regulate genes required for mating, also functions with Tec1p to regulate genes required for pseudohyphal growth
YER155C bem2 GTPase-activating (GAP) protein that regulates Rho1p and has a role in bud emergence and cell cycle-related cytoskeletal reorganization, has a GAP-independent function in the morphogenesis checkpoint, involved in ethanol tolerance
YML048W-A
YML047C PRM6 pheromone-regulated membrane protein
YOR343C
YMR232W fus2 involved in cell fusion
YIL037C PRM2 pheromone-regulated membrane protein
YBR040W FIG1 Protein required for efficient mating
YNL279W PRM1 pheromone-regulated membrane protein
YCR089W FIG2 Protein involved in mating induction
YPL156C PRM4 pheromone-regulated membrane protein
YDR282C
YDL127W PCL2 G1 cyclin
YML046W PRP39 RNA splicing factor associated with U1 snRNP
YFL047W RGD2 GTPase activating protein (GAP) containing a Fes/CIP4 domain, a dishevelled-Egl-10-pleckstrin homology domain, a Rho-GAP domain, and a PEST motif, has similarity to Rga2p

YJL157C FAR1 Inhibitor of Cdc28p-Cln1p and Cdc28p-Cln2p kinase complexes involved in cell cycle arrest for mating
YMR065W KAR5 Coiled-coil membrane protein required for homotypic nuclear fusion
YCL055W KAR4 Regulatory protein required for pheromone induction of karyogamy genes
YLR452C sst2 Regulator of G protein signaling family member that negatively regulates the mating pheromone response pathway by binding to Gpa1p and stimulating its intrinsic GTPase activity, leading to desensitization to pheromone
YNR044W aga1 anchorage subunit of a-agglutinin
YCL027W FUS1 Protein required for cell fusion during mating, contains an SH3 domain, localized to the tip of the mating projection (shmoo)
YNL106C INP52 inositol polyphosphate 5-phosphatase
YNL192W CHS1 chitin synthase 1
YDR309C GIC2 Cdc42 binding protein, involved in bud emergence

Group 8

YMR134W Protein involved in iron metabolism
YKL008C LAC1 Protein required with Lag1p for ceramide synthesis and ER-to-Golgi transport of GPI-anchored proteins
YGR060W ERG25 C-4 sterol methyl oxidase
YML008C erg6 S-adenoslymethionine: delta 24-methyltransferase
YGL001C ERG26 C-3 sterol dehydrogenase
YER044C ERG28 may facilitate protein-protein interactions between the Erg26p dehydrogenase and the Erg27p 3-ketoreductase and/or to tether these enzymes to the ER
YHR190W ERG9 squalene synthetase
YPL028W erg10 acetoacetyl CoA thiolase
YMR202W ERG2 C-8 sterol isomerase
YPL117C idi1 Isopentenyl diphosphate:dimethylallyl diphosphate isomerase (IPP isomerase)
YGR266W
YNR043W MVD1 mevalonate pyrophosphate decarboxylase
YJR047C ANB1 anaerobically expressed form of translation initiation factor eIF-5A
YDR024W FYV1
YEL034W HYP2 Translation initiation factor eIF-5A
YEL033W Protein required for invasive growth and pseudohyphal development
YNL111C CYB5 cytochrome b5
YML075C hmg1 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase isozyme
YMR015C ERG5 cytochrome P450 involved in C-22 denaturation of the ergosterol side-chain
YMR009W Member of the acireductone dioxygenase family
YLR205C HMX1 "Homology to heMe oXygenases"
YJL167W ERG20 Farnesyl diphosphate synthetase (FPP synthetase)
YNR019W ARE2 Acyl-CoA cholesterol acyltransferase (sterol-ester synthetase)

YPL041C
YOR229W WTM2 Transcriptional modulator
YDR044W HEM13 Coproporphyrinogen III oxidase
YMR145C NDE1 Mitochondrial NADH dehydrogenase that catalyzes cytosolic NADH oxidation
YJR048W cyc1 iso-1-cytochrome c
YER011W TIR1 Stress-induced cell wall mannoprotein, member of the seripauperin (PAU) family
YLR056W ERG3 C-5 sterol desaturase
YLL056C
YCR103C
YKL224C Member of the semipauperin (PAU) family; subtelomerically encoded
YIR041W
YLR461W PAU4 member of the seripauperin protein family
YLR309C IMH1
YBR087W RFC5 Subunit 5 of Replication Factor C\; homologous to human RFC 38 kDa subunit
YKR046C Protein possibly involved in respiration
YJL048C
YDL086W Member of the dienelactone hydrolase family and putative carboxymethylenebutenolidase
YLR232W
YER122C GLO3 Zinc-finger-containing protein with similarity to Gcs1p and Sps18p
YPL194W DDC1 DNA damage checkpoint gene
YFR033C QCR6 ubiquinol-cytochrome c oxidoreductase subunit 6 (17 kDa)
YOR065W CYT1 Cytochrome c1
YEL024W RIP1 Rieske iron-sulfur protein of the mitochondrial cytochrome bc1 complex
YBR242W Probable ATP\/GTP-binding protein
YMR220W ERG8 48 kDa Phosphomevalonate kinase
YCL063W VAC17 Protein required for vacuolar segregation, functions as a Myo2p receptor at the vacuole
YLR194C
YOR301W RAX1 Similar to B-chains of the insulin-related hormone superfamily.
YAR003W SWD1 beta transducin domain
YOR377W ATF1 Alcohol acetyltransferase
YGL055W ole1 delta-9-fatty acid desaturase
YML126C ERG13 3-hydroxy-3-methylglutaryl coenzyme A synthase
YLR153C ACS2 acetyl-coenzyme A synthetase
YMR208W erg12 mevalonate kinase
YLL012W
YHR072W ERG7 2,3-oxidosqualene-lanosterol cyclase
YNL156C
YLR326W
YGR235C
YGR234W YHB1 Flavohemoglobin

YMR061W RNA14 component of the cleavage and polyadenylation factor CF I involved in pre-mRNA 3'-end processing
YIR024C GIF1 (putative) involved in cell cycle control
YEL039C cyc7 iso-2-cytochrome c
YLR101C
YLR100W ERG27 3-keto sterol reductase
YNL017C
YDL093W PMT5 dolichyl phosphate-D-mannose:protein O-D-mannosyltransferase
YGL039W
YDR213W UPC2 zinc finger transcription factor of the Zn(2)-Cys(6) binuclear cluster domain type
YGR209C trx2 thioredoxin
YGL160W Member of the Ferric reductase like transmembrane component family, which may bind FAD and move electrons across the plasma membrane
YMR070W MOT3 2 Cys2-His2 zinc fingers at c-terminus, glutamine and asparagine rich
YIR008C PRI1 p48 polypeptide of DNA primase
YOR117W RPT5 26S protease regulatory subunit
YDR155C CPH1 cyclophilin peptidyl-prolyl cis-trans isomerase
YHR004C NEM1 Nuclear Envelope Morphology
YNL280C ERG24 sterol C-14 reductase
YHR191C CTF8 (putative) kinetochore protein
YPR131C NAT3 N-terminal acetyltransferase
YGR178C PBP1 interacts with poly(A)-binding protein
YDR167W TAF25 TFIID subunit
YNR029C
YGL210W YPT32 ras-like GTPase, highly homologous to YPT31
YNR075W COS10 Protein with strong similarity to subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p
YGL261C
YDL133W
YGL056C SDS23 similar to S. pombe sds23
YNL094W APP1 Protein that interacts with Abp1p and Rvs1p, may have a role in the cytoskeleton, involved in the utilization of non-glucose carbon sources, may regulate transcription through the Srb/mediator complex
YNR010W CSE2 Component of RNA polymerase II mediator subcomplex
YGR272C
YJR126C VPS70 Protein involved in Prc1p vacuolar trafficking pathway, contains a PA (protease associated) domain
YLR248W RCK2 Serine\/threonine protein kinase
YGR281W YOR1 ABC transporter
YOR333C
YOL132W
YOR348C PUT4 putative proline-specific permease
YPR150W
YDL174C DLD1 mitochondrial enzyme D-lactate ferricytochrome c oxidoreductase
YBR183W YPC1 alkaline ceramidase with reverse activity

YER145C FTR1 Iron permease
YLR380W CSR1 chs5 spa2 rescue\; isolated as a multicopy suppressor of the lethality of chs5 spa2 double mutant at 37 degrees.
YOR099W KTR1 mannosyltransferase involved in O- and N-linked glycosylation
YOR002W ALG6 glucosyltransferase
YOR321W PMT3 dolichyl phosphate-D-mannose:protein O-D-mannosyltransferase
YDL193W
YDR518W EUG1 Protein disulfide isomerase homolog
YOL013C HRD1 required to degrade misfolded ER lumenal and integral membrane proteins
YNL052W COX5A Cytochrome-c oxidase chain Va
YGL191W COX13 subunit VIa of cytochrome c oxidase, may specifically interact with ATP
YMR038C lys7 copper chaperone for superoxide dismutase Sod1p
YAL028W

Group 9

YFL068W Protein that may bind phosphatidylinositol-4,5-bisphosphate, has similarity to other subtelomerically-encoded proteins including Yhr219p, Yll066p, and Yhl050p
YGR292W MAL12 Maltase
YDR544C
YEL076C Protein with similarity to other subtelomerically-encoded proteins
YEL076C-A
YFL066C member of the subtelomeric Y' family of putative helicase
YBL112C
YHR218W member of the subtelomeric Y' family of putative helicase
YBL111C
YFL067W
YHR219W member of the subtelomeric Y' family of putative helicase
YPL283C YRF1-7 Y'-helicase protein 1
YGR296W YRF1-3 Y'-helicase protein 1
YIL177C member of the subtelomeric Y' family of putative helicase
YNL339C YRF1-6 Y'-helicase protein 1
YML133C member of the subtelomeric Y' family of putative helicase
YLL066C
YHL050C member of the subtelomeric Y' family of putative helicase
YEL077C member of the subtelomeric Y' family of putative helicases
YJL225C member of the subtelomeric Y' family of putative helicase
YER190W YRF1-2 Y'-helicase protein 1
YDR545W YRF1-1 Y'-helicase protein 1
YBL113C member of the subtelomeric Y' family of putative helicase
YLL067C member of the subtelomeric Y' family of putative helicase
YFL064C
YFL065C
YHL049C

YBL109W
YNL338W
YJR115W
YCR107W aad3 Hypothetical aryl-alcohol dehydrogenase (AAD)
YOL165C aad15 Hypothetical aryl-alcohol dehydrogenase (AAD)
YCR106W Transcription regulator
YGR217W CCH1 putative calcium channel

Group 10

YKL175W ZRT3 Vacuolar membrane protein involved in zinc transport and the regulation of zinc storage, transports zinc from an intracellular compartment in response to zinc deficiency
YNR039C ZRG17 zinc-regulated gene
YJL056C ZAP1 Metalloregulatory protein involved in zinc-responsive transcriptional regulation
YGL255W ZRT1 high-affinity zinc transport protein
YAL044C GCV3 H-protein subunit of the glycine cleavage system
YMR189W GCV2 Glycine decarboxylase complex (P-subunit), glycine synthase (P-subunit), Glycine cleavage system (P-subunit)
YDR019C GCV1 glycine cleavage T protein (T subunit of glycine decarboxylase complex
YHR111W UBA4 Ubiquitin-activating enzyme that activates ubiquitin related protein, Urm1p
YCL064C cha1 catabolic serine (threonine) dehydratase
YPL265W DIP5 dicarboxylic amino acid permease
YLR355C ilv5 acetohydroxyacid reductoisomerase
YHR162W
YJL213W Member of the amidohydrolase family, which hydrolyze adenine to form hypoxanthine and ammonia; Adenine catabolism
YJR148W BAT2 Branched-Chain Amino Acid Transaminase
YLR304C aco1 Aconitase, mitochondrial
YML076C WAR1 Transcription factor with a Zn[2]-Cys[6] fungal-type binuclear cluster domain, involved in mediating the weak acid stress response
YNR061C
YMR300C ade4 phosphoribosylpyrophosphate amidotransferase
YMR120C ADE17 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR) transformylase\/IMP cyclohydrolase
YLR058C SHM2 serine hydroxymethyltransferase
YBL029W

Group 11

YML011C
YER083C
YER023W pro3 delta 1-pyrroline-5-carboxylate reductase

YLR209C PNP1 Purine Nucleoside Phosphorylase
YML085C TUB1 alpha-tubulin
YGL214W
YPL113C
YER120W SCS2 involved in inositol metabolism, regulator of INO1 expression
YPL259C APM1 medium subunit of the clathrin-associated protein complex
YDR222W
YPL104W MSD1 Aspartyl-tRNA synthetase, mitochondrial
YMR303C adh2 alcohol dehydrogenase II
YCL032W ste50 STE50
YGR028W MSP1 40 kDa putative membrane-spanning ATPase
YGL219C MMM2
YOR164C
YPR069C SPE3 putrescine aminopropyltransferase (spermidine synthase)
YGL012W ERG4 Sterol C-24 reductase
YCR024C-A PMP1 Proteolipid associated with plasma membrane H(+)-ATPase (Pma1p)
YMR316W DIA1 digs into agar
YOR256C
YKL079W SMY1 kinesin heavy chain homolog
YDR197W CBS2 cytochrome b translational activator; active in mitochondria
YMR257C pet111 translational activator of cytochrome c oxidase subunit II
YCR024C Asn-tRNA synthetase, mitochondrial
YDR494W
YIL093C RSM25 protein of the small subunit of the mitochondrial ribosome
YDR322W MRPL35 Mitochondrial ribosomal protein MRPL35 (YmL35)
YHR059W FYV4
YHR147C MRPL6 Mitochondrial ribosomal protein MRPL6 (YmL6)
YNL315C ATP11 F1FO ATPase assembly protein
YMR188C
YGR021W
YDR462W MRPL28 Mitochondrial ribosomal protein MRPL28 (YmL28)
YLR312W-A MRPL15 Mitochondrial ribosomal protein MRPL15 (YmL15)
YPL215W CBP3 involved in cytochrome-c reductase assembly; active in mitochondria
YPL118W MRP51 Component of small subunit of the mitochondrial ribosome
YGL143C MRF1 Mitochondrial polypeptide chain release factor
YNL252C MRPL17 mitochondrial ribosomal protein of the large subunit
YOR158W pet123 Mitochondrial ribosomal protein of small subunit
YDR337W MRPS28 Mitochondrial ribosomal protein MRPS28 (E. coli S15)
YOL033W MSE1 Mitochondrial glutamyl-tRNA synthetase
YER017C AFG3 ATP-dependent metalloprotease
YJR122W CAF17 component of CCR4 transcriptional complex
YMR293C
YMR283C rit1 Initiator methionine tRNA 2'-O-ribosyl phosphate transferase
YPL072W UBP16 putative deubiquitinating enzyme
YKL155C RSM22

YEL001C
YBR122C MRPL36 Mitochondrial ribosomal protein MRPL36 (YmL36)
YJR103W URA8 CTP synthase
YFR009W GCN20 Member of ATP-binding cassette (ABC) family of proteins
YDL160C DHH1 (putative) DEAD box RNA helicase
YOL026C
YOR185C gsp2 Ran, a GTP-binding protein member of the ras superfamily involved in trafficking through nuclear pores
YDR304C CYP5 Cyclophilin D, Peptidyl-prolyl cis-trans isomerase D
YPR126C
YPR125W
YDR194C MSS116 Mitochondrial DEAD box RNA helicase
YKL207W
YFR044C
YDR212W tcp1 chaperonin subunit alpha
YPL140C MKK2 protein kinase
YLR447C vma6 vacuolar ATPase V0 domain subunit d (36 kDa)
YHR068W DYS1 Deoxyhypusine synthase
YOR335C ALA1 Cytoplasmic alanyl-tRNA synthetase gene
YIL078W THS1 Threonyl-tRNA synthetase, cytoplasmic
YMR023C MSS1 putative mitochondrial GTPase
YHR019C DED81 Asparaginyl-tRNA synthetase
YDR261C exg2 Exo-1,3-b-glucanase
YER087W Putative prolyl-tRNA synthetase
YLR253W
YNL184C
YPL097W MSY1 Tyrosyl-tRNA synthetase, mitochondrial
YDR316W
YOL023W IFM1 mitochondrial initiation factor 2
YOR266W PNT1 involved in pentamidine resistance protein
YHR091C MSR1 Arginyl-tRNA synthetase, mitochondrial
YHR109W CTM1 Cytochrome c methyltransferase
YHR011W DIA4 Seryl-tRNA synthetase
YMR098C
YLR239C LIP2 active in mitochondria
YFR007W
YNL083W
YNR045W pet494 translational activator of cytochrome c oxidase
YER058W PET117 cytochrome c oxidase assembly factor; active in mitochondria
YKL138C MRPL31 15.5 kDa mitochondrial ribosomal protein YmL31
YDR041W RSM10 protein of the small subunit of the mitochondrial ribosome
YJR031C GEA1 GDP\/GTP exchange factor
YOR201C pet56 Ribose methyltransferase for mitochondrial 21S rRNA
YBL107C
YBR251W MRPS5 Probable mitochondrial ribosomal protein S5
YBL090W MRP21 Component of the small subunit of mitochondrial ribosomes

YBR282W MRPL27 Mitochondrial ribosomal protein MRPL27 (YmL27)
YBR120C CBP6 Translational activator of COB mRNA
YBL038W MRPL16 Mitochondrial ribosomal protein MRPL16
YBR262C
YBR192W RIM2 Probable carrier protein, mitochondrial
YOL072W THP1 Protein involved in connecting transcription elongation with mitotic recombination
YOL002C
YML065W ORC1 120-kDa (largest) subunit of origin recognition complex (ORC)\; shows homology to Cdc6p, Cdc18p, and Sir3p and to proteins from K. lactis, S. pombe, and humans
YMR071C
YPL040C ISM1 nuclear encoded mitochondrial isoleucyl-tRNA synthetase
YER078C
YPR063C
YPR062W FCY1 cytosine deaminase
YJR139C hom6 Homoserine dehydrogenase (L-homoserine:NADP oxidoreductase)
YHR026W ppa1 vacuolar ATPase V0 domain subunit c"
YPR036W vma13 vacuolar ATPase V1 domain subunit H (54 kDa)
YKL127W pgm1 phosphoglucomutase, minor isoform
YOR103C OST2 16-kDa, epsilon subunit of oligosaccharyltransferase complex\; 40\% identical to vertebrate DAD1 protein
YNL168C
YMR124W
YFL007W BLM3 Involved in protecting the cell against bleomycin damage
YGL235W
YDR145W TAF61 TFIID subunit
YMR228W mtf1 Mitochondrial RNA polymerase specificity factor
YLR260W LCB5 sphingoid long chain base (LCB) kinase
YOR074C cdc21 Thymidylate synthase
YFL017W-A SMX2 snRNP G protein (the homologue of the human Sm-G)
YNL130C CPT1 sn-1,2-diacylglycerol cholinephosphotransferase
YJL174W kre9 involved in beta-1,6-glucan assembly
YOR122C pfy1 profilin (actin-binding protein)
YLL050C COF1 Cofilin, actin binding and severing protein
YNR035C ARC35 Arp Complex Subunit
YKR001C VPS1 putative GTP-binding protein\; similar to mammalian Mx proteins
YDL100C
YJR087W
YHR061C GIC1 interacts with GTPase, involved in bud emergence
YPR047W MSF1 alpha subunit of yeast mitochondrial phenylalanyl-tRNA synthetase
YLR382C nam2 mitochondrial leucyl tRNA synthetase
YDR477W snf1 protein serine\/threonine kinase
YDR454C GUK1 guanylate kinase

YJL002C OST1 Oligosaccharyltransferase alpha subunit, member of a complex of eight ER proteins that transfers core oligosaccharide from dolichol carrier to Asn-X-Ser/Thr motif
YEL002C WBP1 oligosaccharyl transferase glycoprotein complex, beta subunit
YJL192C
YHR133C
YNL004W HRB1 hypothetical RNA-binding protein
YDL084W SUB2 RNA helicase□
YCL001W RER1 Golgi protein involved in retention of ER proteins
YMR302C prp12 Integral membrane mitochondrial protein
YMR166C
YPL085W SEC16 vesicle coat component
YGL215W CLG1 cyclin-like protein that interacts with Pho85
YMR099C
YOL090W msh2 MutS homolog encoding major mismatch repair activity in mitosis and meiosis.□
YCL052C PBN1 Protease B, nonderepressible form
YIR001C SGN1 Contains one RNA recognition (RRM) domain
YBR227C MCX1 Mitochondrial ATP-binding protein, similar to ClpX
YBR276C PPS1 dual specificity protein phosphatase
YBL091C MAP2 methionine aminopeptidase 2
YBL033C rib1 GTP cyclohydrolase II
YLR257W
YDR313C PIB1 phosphatidylinositol(3)-phosphate binding protein
YGR106C
YGL200C EMP24 type I transmembrane protein, component of COPII-coated, ER-derived transport vesicles
YJR088C
YJR142W
YEL051W VMA8 vacuolar ATPase V1 domain subunit D
YGL106W MLC1 light chain for myosin Myo2p
YHR176W Dimethylaniline monoxygenase
YDL040C nat1 N-terminal acetyltransferase
YDL001W
YPR008W HAA1
YKR021W
YDL190C UFD2 ubiquitin fusion degradation protein
YDL185W TFP1 vacuolar ATPase V1 domain subunit A (69 kDa)
YGR132C PHB1 mitochondrial protein, prohibitin homolog\; similar to S. cerevisiae Phb2p
YDR153C
YPL260W
YDL178W AIP2 D-Lactate Dehydrogenase (Cytochrome), mitochondrial
YJL062W LAS21 putative membrane protein, a member of the major facilitator super family
YHL017W Probable transmembrane protein PTM1

YNL229C ure2 transcriptional regulator, putative glutathione transferase
YOL051W gal11 Component of the RNA polymerase II holoenzyme complex, positive and negative transcriptional regulator of genes involved in mating-type specialization
YPL184C
YIL115C NUP159 159-kDa nucleoporin with coiled-coil domain and repeated motifs typical of nucleoporins
YIL137C
YNL262W pol2 DNA polymerase II
YML103C NUP188 Nucleoporin
YOR153W pdr5 multidrug resistance transporter
YPL231W FAS2 alpha subunit of fatty acid synthase
YML120C NDI1 NADH dehydrogenase (ubiquinone)
YJR080C
YDR170C sec7 Guanine nucleotide exchange protein for ARF
YNL085W mkt1 retroviral protease signature protein
YBL066C SEF1 putative transcription factor
YDR524C AGE1 ARF GAP with effector function(s)
YKL170W MRPL38 mitochondrial ribosomal protein L14
YKL169C
YKL053C-A
YMR267W ppa2 mitochondrial inorganic pyrophosphatase
YPL073C
YMR193W MRPL24 Mitochondrial ribosomal protein MRPL24 (YmL24)
YKR006C MRPL13 mitochondrial ribosomal protein YmL13
YOL071W
YER182W
YML050W
YDR433W KRE22
YBL032W
YBR011C ipp1 Inorganic pyrophosphatase
YBR243C alg7 UDP-N-acetyl-glucosamine-1-P transferase (GPT)
YJL173C RFA3 subunit 3 of replication factor-A
YCL011C GBP2 Protein with RNA recognition motifs
YDR260C SWM1 Spore Wall Maturation 1
YOL042W NGL1 Protein with possible DNAse/RNAse function
YOL003C
YER156C
YFR016C
YKL003C mrp17 Mitochondrial ribosomal protein MRP17
YDR419W RAD30 similar to E. coli DinB and S. cerevisiae REV1
YDL107W mss2 cox1 pre-mRNA splicing factor
YPR100W
YLR204W QRI5
YDR237W MRPL7 Mitochondrial ribosomal protein MRPL7 (YmL7)
YDR116C
YJL096W MRPL49 mitochondrial ribosomal protein of the large subunit

YCR003W MRPL32 Mitochondrial ribosomal protein MRPL32 (YmL32)
YDL202W MRPL11 Mitochondrial ribosomal protein MRPL11 (YmL11)
YNL177C falls in mitochondrial expression cluster, Hughes et al.
YNL005C MRP7 Mitochondrial ribosomal protein MRP7 (YmL2) (E. coli L27)
YEL050C RML2 mitochondrial ribosomal protein L2 of the large subunit
YNL137C nam9 putative mitochondrial S4 ribosomal protein
YOR150W MRPL23 mitochondrial ribosomal protein of the large subunit
YGR165W falls in mitochondrial expression cluster, Hughes et al.
YPR099C
YBR146W MRPS9 Probable mitochondrial ribosomal protein S9
YIL098C FMC1 Formation of Mitochondrial Cytochromes 1
YIL157C
YPR133W-A TOM5 Membrane protein involved in protein translocation to the mitochondria
YGL069C
YOR286W
YER028C
YPR157W
YKL117W SBA1 Hsp90 (Ninety) Associated Co-chaperone
YFL038C YPT1 Ras-like GTP-binding protein\; most similar to mammalian Rab1A protein
YNL086W
YNL009W IDP3 peroxisomal NADP-dependent isocitrate dehydrogenase
YPR045C
YJR138W IML1
YGR093W
YFR024C-A
YFR024C
YJR065C ARP3 involved in the function of the actin cytoskeleton
YPL249C
YOR124C UBP2 Ubiquitin-specific protease
YDL126C CDC48 microsomal ATPase
YDR457W TOM1 hect-domain-containing protein, containing kinase motifs\; similar to Rsp5
YGL173C KEM1 cytoplasmic 5'-to-3' exonuclease.
YHR099W TRA1 ATM\/Mec1\/TOR1+2-related
YLR118C
YKL068W NUP100 Nuclear pore complex protein homologous to Nup116p
YHR120W msh1 mutS homolog involved in mitochondrial DNA repair
YDR368W YPR1 similar to aldo-keto reductase
YOR205C
YLR183C TOS4
YEL047C
YNL045W
YPL029W suv3 putative ATP-dependent helicase
YDR393W SHE9 similar to Arabidopsis Cip1, lethal when overexpressed

YJL004C SYS1 Multicopy suppressor of ypt6 null mutation
YML112W CTK3 CTD kinase-I gamma subunit
YHR038W FIL1 Putative mitochondrial ribosome recycling factor
YMR178W
YDL122W UBP1 Ubiquitin-specific protease
YIL128W MET18 regulator of TFIIH
YML087C
YMR287C MSU1 component of 3'-5'exonuclease complex
YLR067C PET309 involved in COX1 mRNA stability
YJL209W CBP1 involved in control of mRNA stablility
YGL002W ERP6 p24 protein involved in membrane trafficking
YOR334W MRS2 splicing factor
YDR430C
YGL129C RSM23 protein of the small subunit of the mitochondrial ribosome
YGL107C
YGR084C MRP13 35 kDa mitochondrial ribosomal small subunit protein
YDR175C RSM24 protein of the small subunit of the mitochondrial ribosome
YOR354C MSC6
YIL070C MAM33 Mitochondrial protein involved in respiration
YMR024W MRPL3 Mitochondrial ribosomal protein MRPL3 (YmL3)
YHL004W MRP4 mitochondrial ribosomal protein, homologous to E. coli ribosomal protein S2, component of the 37 S subunit of mitochondrial ribosomes
YDR375C BCS1 Mitochondrial ATPase (AAA family)
YPL172C COX10 Putative farnesyl transferase required for heme A synthesis
YMR158W
YLR069C MEF1 mitochondrial elongation factor G-like protein
YNL284C MRPL10 Mitochondrial ribosomal protein MRPL10 (YmL10)
YNR040W
YLR439W MRPL4 Mitochondrial 60S ribosomal protein L4
YIR021W mrs1 mitochondrial RNA splicing
YDL045W-A MRP10 homologous to Yml37p, component of the 37 S subunit of mitochondrial ribosomes
YJL112W MDV1 Mitochondrial DiVision
YGR169C
YCR071C IMG2 similar to Drosophila gonadal protein Z600\; involved in mitochondrial DNA maitenance
YKL087C CYT2 cytochrome c1 heme lyase
YHR116W falls in mitochondrial expression cluster, Hughes et al.
YJL063C MRPL8 Mitochondrial ribosomal protein MRPL8 (YmL8) (E. coli L17)
YDR347W mrp1 37 kDa mitochondrial ribosomal protein
YDR493W
YKL167C MRP49 16 kDa mitochondrial ribosomal large subunit protein
YGR220C MRPL9 Mitochondrial ribosomal protein MRPL9 (YmL9) (E. coli L3) (human MRL3)
YER172C BRR2 putative ATP-dependent RNA helicase
YGR218W CRM1 omosome region maintenance protein

YGL195W GCN1 translational activator of GCN4 through activation of GCN2 in response to starvation
YBR129C OPY1 involved in mating pathway
YBL080C PET112 62-kDa protein
YBR003W COQ1 hexaprenyl pyrophosphate synthetase
YIL022W TIM44 48.8 kDa protein involved in mitochondrial protein import
YHR048W
YML117W
YGL236C MTO1 Mitochondrial Translation Optimization\; Strong similarity to E. coli GidA
YMR092C AIP1 actin cortical patch component
YGR147C NAT2 N alpha-acetyltransferase that acts on methionine termini
YDR168W cdc37 (putative) chaperone, involved in spindle pole body duplication and passage through START
YJR086W ste18 gamma subunit of G protein coupled to mating factor receptors
YPL225W
YIR007W
YMR003W
YLR225C
YCR036W RBK1 ribokinase
YBL009W
YBR041W FAT1 Fatty acid transporter
YKL157W APE2 aminopeptidase yscII
YJL068C Esterase, may be involved in conversion of formaldehyde to S-formylglutathione
YFR047C
YEL015W
YML080W DUS1 tRNA dihydrouridine synthase
YBR053C
YPR180W AOS1 Similar to ubiquitin activating proteins
YKR095W mlp1 colied-coil protein (putative), similar to myosin and TPR
YNL204C SPS18 Transcription factor
YGR241C YAP1802 member of clathrin assembly polypeptide AP180 family
YER032W FIR1 Putative participant in 3' mRNA processing
YDR388W rvs167 (putative) cytoskeletal protein
YCR088W ABP1 Actin binding protein
YDR141C DOP1 homolog of Emericella nidulans developmental regulatory gene, dopey (dopA).
YHR074W QNS1
YOL050C
YCL024W KCC4 protein kinase related to S. pombe Nim1p
YJR113C RSM7 protein of the small subunit of the mitochondrial ribosome
YCR012W PGK1 3-phosphoglycerate kinase
YKL153W
YKL152C gpm1 Phosphoglycerate mutase
YDR206W EBS1 similar to Est1, which is a putative component of telomerase

YGR219W
YMR286W MRPL33 Mitochondrial ribosomal protein MRPL33 (YmL33) (E. coli L30)
YNR020C
YER050C RSM18 protein of the small subunit of the mitochondrial ribosome
YOR159C SME1 homologue of human E core protein; Required for pre-mRNA splicing, cap modification and U1, U2, U4 and U5 snRNA stability
YOR020C HSP10 10 kDa mitochondrial heat shock protein
YNL122C
YJR101W RSM26 protein of the small subunit of the mitochondrial ribosome
YGR215W RSM27 protein of the small subunit of the mitochondrial ribosome
YGL068W
YLR008C
YKR085C MRPL20 22.3 kDa mitochondrial ribosomal large subunit protein YmL20\; homologous to L17 of E. coli
YBR110W alg1 beta-1,4-mannosyltransferase
YBR017C KAP104 karyopherin beta 2, yeast transportin
YBR127C VMA2 vacuolar ATPase V1 domain subunit B (60 kDa)
YER105C NUP157 Nucleoporin similar to Nup157p and to mammalian Nup155p
YDL145C COP1 alpha subunit of the coatamer complex\; gamma-alpha-COP
YNL288W CAF40 CCR4 associated factor 40 kDa
YOR043W whi2 involved in growth regulation
YGR171C MSM1 mitochondrial methionyl-tRNA synthetase
YGR076C MRPL25 Mitochondrial ribosomal protein MRPL25 (YmL25)
YGR216C gpi1 involved in first step of GPI (N-acetylglucosaminylphosphatidylinositol) anchor synthesis
YAL021C ccr4 95 kDa containng leucine rich tandem repeats
YNR052C POP2 Putative transcription factor
YCL051W LRE1 involved in laminarase resistance
YMR157C falls in mitochondrial expression cluster, Hughes et al.
YJL180C ATP12 (putative) involved F1F0-ATPase complex assembly; active in mitochondria
YNL081C Putative mitochondrial ribosomal protein of the small subunit, has similarity to ribosomal protein S13 of prokaryotes and plant mitochondria
YGR062C COX18 required for mitochondrial cytochrome oxidase activity
YOR200W
YPR024W YME1 Mitochondrial member of the CDC48\/PAS1\/SEC18 family of ATPases
YPL103C
YPL105C
YNR033W ABZ1 para-aminobenzoate synthase, PABA synthase
YJL207C
YPR011C
YPR067W ISA2 Iron Sulfur Assembly -- IscA\/NifA homolog

Group 12

YBR177C EHT1 alcohol acyl transferase
YOR353C
YML110C COQ5 Putative mitochondrial C-methyltransferase
YNL133C FYV6
YDR320C SWA2 Auxilin-like protein
YLL028W TPO1 polyamine transport protein
YOL108C INO4 Basic helix-loop-helix (bHLH) transcription factor involved in activation of phospholipid synthetic genes
YKL072W STB6 involved in transcription
YHR044C DOG1 2-deoxyglucose-6-phosphate phosphatase
YML088W UFO1 F-box protein
YDR088C SLU7 involved in mRNA splicing
YJR074W MOG1 nuclear protein that interacts with GTP-Gsp1p
YHR057C CYP2 Peptidylprolyl isomerase (cyclophilin) ER or secreted
YKL179C kinesin-like protein
YBR198C TAF90 Probable transcription-associated factor protein, probable -transducin type
YDR406W PDR15 probable multidrug resistance transporter
YNL234W Hemoprotein with similarity to mammalian globins
YFL015C
YGL043W dst1 RNA polymerase II elongation factor
YDR138W hpr1 involved in mitosis, recombination\; similar to TOP1 across 2 regions
YKR093W PTR2 Peptide transporter
YHR139C-A
YER001W mnn1 Alpha-1,3-mannosyltransferase
YER095W rad51 Protein that stimulates pairing and strand-exchange between homologous single-stranded and double-stranded DNA, functionally similar to E. coli RecA protein
YJL186W MNN5 putative Golgi alpha-1,2-mannosyltransferase
YOR173W
YDR100W
YOR161C
YCL029C BIK1 Microtubule-binding protein
YKR018C
YAL008W FUN14
YGR142W BTN2 Gene\/protein whose expression is elevated in a btn1 minus\/Btn1p lacking yeast strain.
YKL074C MUD2 involved in early pre-mRNA splicing
YDL019C OSH2 OSbp Homologue (OSBP stands for Oxysterol binding protein)
YER176W ECM32 DNA Helicase I
YOL096C COQ3 3,4-dihydroxy-5-hexaprenylbenzoate methyltransferase
YGL071W RCS1 Putative transcription factor that binds the consensus site PyPuCACCCPu
YGR134W CAF130 CCR4 associated factor 130 kDa
YDR056C

YDR055W PST1 The gene product has been detected among the proteins secreted by regenerating protoplasts
YPR145W ASN1 asparagine synthetase
YGR124W ASN2 asparagine synthetase
YKR019C IRS4 involved in rDNA silencing
YMR153W NUP53 Component of karyopherin docking complex of the nuclear pore complex
YER181C
YLR391W-A
YGR042W
YML074C FPR3 Prolyl cis-trans isomerase, also called proline rotamase or peptidylprolyl cis-trans isomerase (PPIase)
YKL066W
YBR261C
YJL042W MHP1 Putative microtubule-associated protein (MAP)
YML070W DAK1 putative dihydroxyacetone kinase
YER062C HOR2 DL-glycerol-3-phosphatase
YDR486C VPS60
YAL060W
YPL230W USV1 Up in StarVation
YOL153C
YGR248W SOL4 similar to SOL3
YGR088W CTT1 cytoplasmic catalase T
YML128C MSC1
YDL204W RTN2 Member of the reticulon family, which are proteins that associate with the endoplasmic reticulum
YIL136W OM45 45-kDa mitochondrial outer membrane protein
YDL223C HBT1 Target of Hub1p ubiquitination, involved in bud site selection and shmoo formation
YDL222C Protein possibly involved in sphingolipid metabolism, has similarity to Sur7p
YOL053C-A
YDR070C Phosphatidylinositol-4,5-bisphosphate binding
YER054C GIP2 (putative) regulator of Glc7, a PP1 family protein phosphatase
YAL061W
YLR178C tfs1 CDC25-dependent nutrient- and ammonia-response cell cycle regulator
YPR184W GDB1 the enzyme that debranches the glycogen having a glucanotranferase + 1-6amyloglucosidase activity
YMR250W GAD1 glutamate decarboxylase
YER150W SPI1 similar to Sed1\; highly expressed in stationary phase
YPL186C
YKL091C
YMR251W-A HOR7 hyperosmolarity-responsive gene
YDR300C pro1 gamma-glutamyl kinase
YMR105C pgm2 Phosphoglucomutase
YML100W tsl1 123 kD regulatory subunit of trehalose-6-phosphate synthase\/phosphatase complex\; homologous to TPS3 gene product

YDR074W TPS2 Trehalose-6-phosphate phosphatase
YBR126C TPS1 56 kD synthase subunit of trehalose-6-phosphate synthase\/phosphatase complex
YHR104W GRE3 a keto-aldose reductase
YPR026W ATH1 Acid trehalase
YJL161W
YDL021W GPM2 phosphoglycerate mutase, involved in glycolysis
YMR090W
YJR096W
YKL151C
YIL113W
YHR016C YSC84 SH3 domain in C-terminus
YBR169C SSE2 HSP70 family member, highly homologous to Sse1p
YLR251W
YNL200C
YGR086C
YPR149W NCE102 Involved in secretion of proteins that lack classical secretory signal sequences
YPL004C
YFL014W HSP12 12 kDa heat shock protein
YER097W
YMR254C
YMR031W-A
YPR091C
YOR086C
YNL160W YGP1 YGP1 encodes gp37, a glycoprotein synthesized in response to nutrient limitation which is homologous to the sporulation-specific SPS100 gene
YOR134W BAG7 GTPase activating protein (GAP)
YKL163W PIR3 Protein containing tandem internal repeats
YIL160C pot1 peroxisomal 3-oxoacyl CoA thiolase
YPL223C GRE1 hydrophilic polypeptide with no known homology
YGR043C
YBR117C TKL2 transketolase, homologous to tkl1
YGR052W Serine/threonine protein kinase of unknown function
YHR080C
YPL161C BEM4 Involved in polarity establishment and bud emergence
YHR096C HXT5 hexose transporter
YMR175W SIP18 Salt-Induced Protein of 18 kDa
YIL101C XBP1 transcriptional repressor
YFR043C
YDR533C Member of the DJ-1 or PfpI family; Phospholipid binding
YKL026C GPX1 Glutathione peroxidase paralogue
YMR170C ALD2 Cytosolic aldeyhde dehydrogenase
YNL195C
YGR201C
YBR001C NTH2 Neutral trehalase, highly homologous to Nth1p

YOL048C
YKL036C
YKL035W UGP1 Uridinephosphoglucose pyrophosphorylase
YGR008C STF2 ATPase stabilizing factor
YPR160W GPH1 Glycogen phosphorylase
YEL011W glc3 1,4-glucan-6-(1,4-glucano)-transferase
YMR196W
YGL037C PNC1 pyrazinamidase and nicotinamidase
YMR252C
YER121W
YBR285W
YNR014W
YJR085C
YMR031C
YOR166C
YLR070C
YJL185C
YGR143W skn1 encodes a predicted type II membrane protein highly homologous to Kre6p
YGR230W BNS1 similar to Spo12, involved in sporulation
YOR317W FAA1 long chain fatty acyl:CoA synthetase
YGL218W
YJL206C-A
YDR349C YPS7 GPI-anchored aspartic protease
YBR296C PHO89 Probable Na+\/Pi symporter
YKL097W-A
YOL047C
YKL051W
YNR036C
YKL085W MDH1 mitochondrial malate dehydrogenase
YDR148C KGD2 dihydrolipoyl transsuccinylase component of alpha-ketoglutarate dehydrogenase complex in mitochondria
YDL181W INH1 ATPase inhibitor
YDL173W
YKL150W MCR1 NADH-cytochrome b5 reductase
YIR037W HYR1 putative glutathione-peroxidase
YCR004C YCP4 FMN-binding protein
YCL040W GLK1 Glucokinase
YOL130W ALR1 putative ion transporter
YPL014W
YKL062W MSN4 zinc finger protein
YKL100C
YMR272C SCS7 desaturase\/hydroxylase enzyme
YNL055C POR1 Outer mitochondrial membrane porin (voltage-dependent anion channel, or VDAC)
YIR038C GTT1 Glutathione transferase

YDR032C PST2 Protoplasts-SecreTed protein\; the gene product was detected among the proteins secreted by regenerating protoplasts
YPL154C pep4 vacuolar proteinase A
YHR087W
YOR152C
YOR262W
YDR343C HXT6 Hexose transporter
YDR342C HXT7 Hexose transporter
YJL214W HXT8 hexose permease
YHR092C hxt4 High-affinity glucose transporter
YKL162C

Group 13

YPL075W GCR1 trans-acting positive regulator of the enolase and glyceraldehyde-3-phosphate dehydrogenase gene families
YML002W
YJR121W ATP2 F(1)F(0)-ATPase complex beta subunit, mitochondrial
YJR120W Protein required for respiratory growth, also has a role in stress resistance and mating
YBR039W ATP3 gamma subunit of mitochondrial ATP synthase
YBL045C COR1 44 kDa core protein of yeast coenzyme QH2 cytochrome c reductase
YBL100C
YKL016C ATP7 ATP synthase d subunit
YJR077C MIR1 Phosphate transporter of the mitochondrial carrier (MCF) family
YDR298C ATP5 ATP synthase subunit 5\; oligomycin sensitivity-conferring protein
YLR168C MSF1' possibly involved in intramitochondrial sorting
YLR294C
YPR020W ATP20 Subunit g homolog of ATP synthase